

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
2       80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

My code:

```
1  
2 y_mean <- mean(y) #finding the mean  
3 y_sd <- sd(y) #finding the standard deviation  
4 y_length <- length(y) #finding the length  
5 y_se <- sd(y)/sqrt(y_length) #finding the standard error
```

```

6 t90 <- qt(((1-0.9)/2), df = y_length-1, lower.tail = FALSE) #finding crit
  value
7
8 lower90 <- y_mean - (t90 * y_se) #finding lower CI number
9 upper90 <- y_mean + (t90 * y_se) #finding upper CI number
10 confint90 <- c(lower90, upper90) #making CI a vector to read
11
12 confint90

```

```
[1] 93.95993 102.92007
```

There is a 90% confidence that the true value of the average IQ of students at the school falls between 93.96 and 102.92.

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

ASSUMPTIONS

Random sample, Quantitative data, Normally distributed population

HYPOTHESIS

NULL Hypothesis: The average IQ at the school is less than or equal to the national average of 100.

Alternative Hypothesis: The average IQ at the school is greater than the national average of 100.

NOTE: This is a one sided test

CALCULATE TEST STATISTIC

Test Statistic = (Sample Statistic - Hypothesized Parameter)/Standard Error

my code:

```
1 y_ts <- (y_mean - 100) / y_se
```

CALCULATE A P-VALUE

pt((ts), df = n-1, lower.tail=F)

One sided test doesn't need abs() or * 2

my code:

```
1 y_pval <- pt((y_ts), df = y_length-1, lower.tail=FALSE)
2 y_pval
```

[1] 0.7215383

CONCLUSION

$\alpha = 0.05$

$p = 0.72$ which is greater than 0.05

We fail to reject the NULL Hypothesis because the p-value is greater than 0.05. The sample does not give evidence that the schools average IQ is higher than 100.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

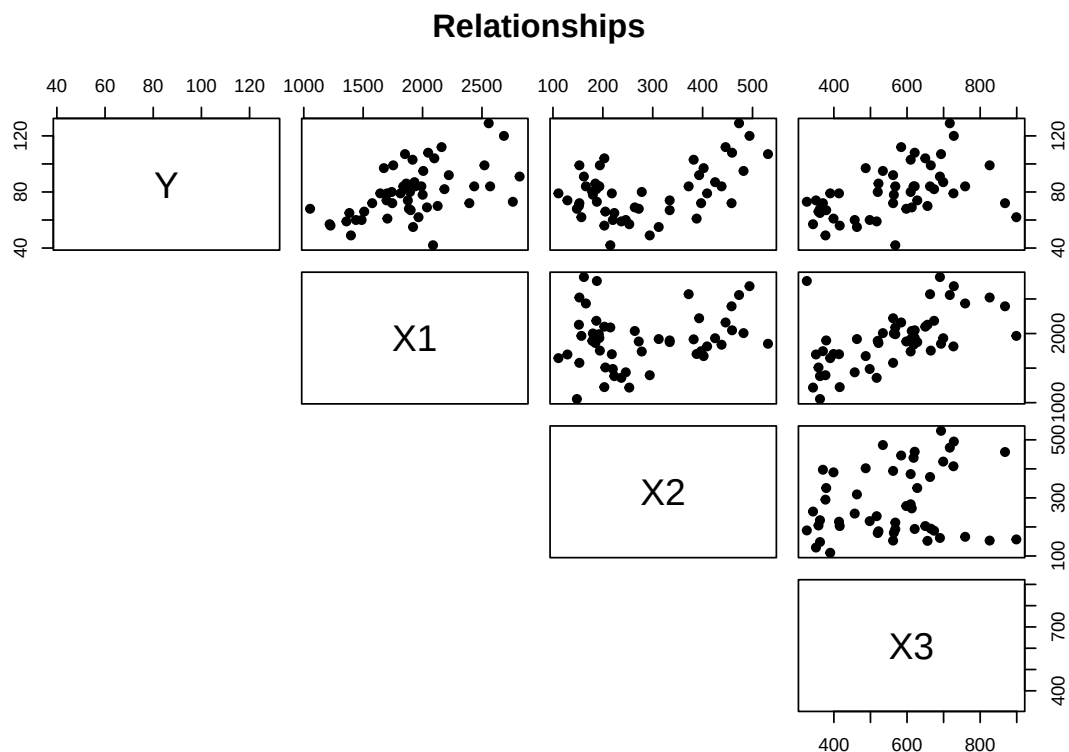
State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

- **2.1** Please plot the relationships among Y , $X1$, $X2$, and $X3$? What are the correlations among them (you just need to describe the graph and the relationships among them)?
my code:

```
1 #using the pairs function to make a graph with multiple overlaying
  variables.
2 pairs(~ Y + X1 + X2 + X3, data = expenditure ,
3     pch = 19,      #plotting with dots
4     main = "Relationships", #giving a title
5     lower.panel = NULL)    #removing redundant graphs
```



```
1 cor(expenditure[, c("Y", "X1", "X2", "X3")]) #getting the correlating
  numbers
```

	Y	X1	X2	X3
Y	1.0000000	0.5317212	0.4482876	0.4636787
X1	0.5317212	1.0000000	0.2056101	0.5952504
X2	0.4482876	0.2056101	1.0000000	0.2210149
X3	0.4636787	0.5952504	0.2210149	1.0000000

my interpretation of the graphs:

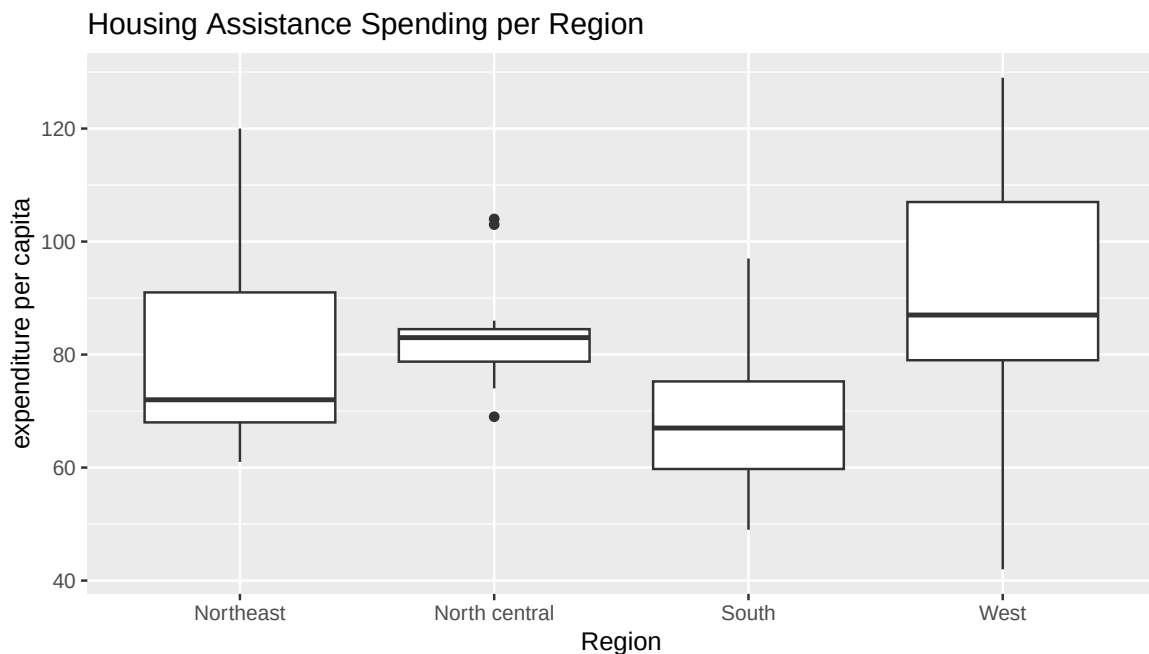
Per capita expenditure on shelters/housing assistance in state (Y) displays a moderate positive correlation with all three variables of X. These X values being: per capita personal income in state (X1), Number of residents per 100,000 that are "financially insecure" in state (X2), and Number of people per thousand residing in urban areas in state (X3). The relationship between Y and X1 has the most clear moderately positive correlation, while both the relationships between Y and X2 and Y and X3 have a moderately positive correlation with Y and X3 being slightly more strongly correlated. The correlations between X2 and X1 as well as X2 and X3 both have a weak positive correlation. The strongest correlation of all the plots is between X3 and X1 which

shows a moderate positive correlation. There is one clear outlier that may be pulling the relationship away from a stronger correlation.

- **2.2** Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

my code:

```
1 #factoring so that I can label my X axis bars and make a boxplot
2 expenditure$Region_fac <- factor(expenditure$Region,
3   labels = c("Northeast", "North central",
4             "South", "West"))
5 #making a boxplot with my regions and Y
6 boxplotvar <- ggplot(expenditure, aes(Region_fac, Y)) +
7   geom_boxplot() +
8   labs(title = "Housing Assistance Spending per Region", #main title
9        x = "Region", y = "expenditure per capita") #giving axis labels
```



```
1 #Finding the means so that I can compare the averages, as the boxplot
   shows the medians:
2 Northeast_mean <- mean(expenditure$Y[expenditure$Region == 1])
3 North_central_mean <- mean(expenditure$Y[expenditure$Region == 2])
4 South_mean <- mean(expenditure$Y[expenditure$Region == 3])
5 West_mean <- mean(expenditure$Y[expenditure$Region == 4])
6 Region_means <- c(Northeast=Northeast_mean, North_central=North_central_
   mean, South=South_mean, West=West_mean) #making a matrix to read
7 Region_means
```

Northeast	North_central	South	West
79.44444	83.91667	69.18750	88.30769

my answer:

On average the West had the highest per capita expenditure on housing assistance.

- **2.3** Please plot the relationship between Y and $X1$? Describe this graph and the relationship.

my code:

```
1 plot(expenditure$X1, expenditure$Y, #plotting X1 and Y
2       main = "Income vs Expenditure", #giving a title
3       xlab = "Per capita personal income", #labelling X axis
4       ylab = "Per capita expenditure on housing assistance") #labelling Y
5       axis
6 fit <- lm(expenditure$Y ~ expenditure$X1) # added a linear regression
7       line
8 abline(fit, col = "red", lwd = 2)
```



```
1 cor(expenditure$Y, expenditure$X1) #getting the correlating number
```

```
[1] 0.5317212
```

my answer:

The graph shows that the per capita expenditure on shelters/housing assistance in state has a moderate positive correlation with per capita personal income in state. The points on the graph appear to begin landing further away from their predicted value more frequently as the values on the x and y axis get larger. A linear regression line was added to better visualize this.

2.3.5 Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

my code:

```
1 plot(expenditure$X1, expenditure$Y, # same code as before
2       main = "Income vs Expenditure per Region", # changed main title to
       reflect
3       xlab = "Per capita personal income",          # the addition of
       Regions
4       ylab = "Per capita expenditure on housing assistance",
5       col = expenditure$Region,                    # change the color of each point per
       region
6       pch = expenditure$Region)                    # change the symbol of each point
       per region
7 legend("topleft",                                # added a legend to understand colors and
       symbols
8       legend = c("Northeast", "North central", "South", "West"),
9       col = 1:4,    # assigned the correct colors to each region in
       legend
10      pch = 1:4)    # assigned the correct symbol to each region in
       legend
11 # same linear regression line but I changed the color as one of the
       regions is now red:
12 fit <- lm(expenditure$Y ~ expenditure$X1)
13 abline(fit, col = "orange", lwd = 2)
```

Income vs Expenditure per Region

